

# Nickole Andrea Villabona

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## RESEARCH INTERESTS

Microbial ecology • Evolutionary ecology • Microbial evolution • Genomics • Biology of symbiosis • Tropical ecology • Bees • Bioinformatics

## EDUCATION

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**Ph.D. Candidate in Ecology and Evolutionary Biology** Fall 2022 – Present  
*University of California, Irvine, USA*

**Master of Science in Biological Sciences** Fall 2022 – Dec 2025  
*University of California, Irvine, USA*

**Bachelor of Science (BS) in Biology** May 2022  
*Universidad de los Andes, Bogotá, Colombia*

**Bachelor of Science (BS) in Microbiology** May 2022  
*Universidad de los Andes, Bogotá, Colombia*

**Minor in Narrative Media** May 2022  
*Universidad de los Andes, Bogotá, Colombia*

## PEER-REVIEWED PUBLICATIONS

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**Villabona, N.**, Moran, N., Hammer, T., & Reyes, A. (2023). Conserved, yet disruption-prone, gut microbiomes in neotropical bumblebees. *mSphere*, e0013923. <https://doi.org/10.1128/msphere.00139-23>

Mahecha, L., **Villabona, N.**, Sierra, L., Ocampo, D., & Laverde-R, O. (2018). The Andean Cock-of-the-rock (*Rupicola peruvianus*) is a frugivorous bird predator. *The Wilson Journal of Ornithology*, 130(2), 558–560.

### Pre-prints (not peer-reviewed)

**Villabona Buitrago, N. A.**, Vazquez, A., Schlauch, J., Rasmussen, C., Joyce, R., & Hammer, T. J. (2026). Gain and loss of gut symbionts during stingless bee diversification is linked to host group size. *bioRxiv* 2026.01.16.699994. <https://doi.org/10.64898/2026.01.16.699994>

## RESEARCH EXPERIENCE

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**Ph.D. Candidate — Hammer Lab** Fall 2022 – Present  
*Dept. of Ecology and Evolutionary Biology, UC Irvine, USA*

- Designed and led projects from sampling to manuscript publication using genomic approaches (16S rRNA, metagenomics) to study host-symbiont dynamics in bees.
- Developed bioinformatics pipelines for amplicon and metagenomic data analysis.

**Summer Undergraduate Research — SUPER Program** Jun 2019 – Aug 2019  
*University of Texas at Austin, Austin, Texas*

- Worked under Dr. Tobin Hammer and Dr. Nancy Moran. Sequenced bumblebee gut microbiomes from Neotropical specimens using molecular lab techniques.

**Undergraduate Research Assistant** Oct 2018 – May 2022  
*BCEM Lab, Universidad de Los Andes, Bogotá, Colombia*

- Research under Dr. Alejandro Reyes. Developed bioinformatics and microbial ecology skills and collaborated with Master's and PhD students.

**Undergraduate Research Assistant** Feb 2018 – Jan 2020  
*LAZOE Lab, Universidad de Los Andes, Bogotá, Colombia*

- Research under Dr. Emilio Realpe on arthropod phylogenetics, ecology, and evolution.

## PROFESSIONAL EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Research Consultant — Microbial Ecology &amp; Bioinformatics</b><br><i>CorpoGen — Research and Biotechnology, Colombia</i> | Jul 2021 – Aug 2022 |
| <b>Marketing and Social Media Manager</b><br><i>Biome Acuity — Microbiome Service Company</i>                                 | Apr 2022 – Oct 2022 |
| <b>Research Assistant</b><br><i>Vice-Chancellor - Office for Research, Universidad de Los Andes</i>                           | Sep 2019 – Jul 2020 |
| <b>Graduate Student Researcher</b><br><i>Dept. of Ecology and Evolutionary Biology, UC Irvine</i>                             | Jan 2023 - Present  |

## TEACHING

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|-------------------------------------------------------------------------------------|------------------------|
| <b>Teaching Assistant — Dept. of Ecology and Evolutionary Biology, UC Irvine</b>    | Apr 2023 – Winter 2026 |
| <b>Teaching Assistant — Principles of Ecology, Universidad de Los Andes</b>         | Aug 2019 – Dec 2019    |
| <b>Teaching Assistant — Invertebrate Biology, Universidad de Los Andes</b>          | Jan 2018 – May 2019    |
| <b>QIIME2 Workshop Instructor — ISME Latin American Congress, Bogotá</b>            | Jul 2021               |
| <b>Microbial Ecology Workshop for Undergraduates (created &amp; led), UC Irvine</b> | Winter 2026 – Present  |

## PRESENTATIONS

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- 2025.** *Winter EEB Graduate Symposium.* Irvine, CA. Group size mediates maintenance of specialized gut symbionts across stingless bee diversification. *Poster presentation.*
- 2024.** *SoCal Microbiome Symposium.* Irvine, CA. Group size mediates maintenance of specialized gut symbionts across stingless bee diversification. *Poster presentation.*
- 2024.** *ISME19 International Symposium on Microbial Ecology.* Cape Town, South Africa. Group size mediates maintenance of specialized gut symbionts across stingless bee diversification. *Poster presentation — ISME Early Career Researcher Poster Award*
- 2023.** *XII Congreso Mesoamericano de Abejas Nativas.* Ciudad Guzmán, Mexico. Abejorros del Neotrópico presentan una microbiota conservada, pero también propensa a alteraciones. *Oral presentation.*
- 2023.** *Yosemite Symbiosis Workshop XI.* Yosemite National Park, USA. Neotropical bumblebee gut microbiota are host conserved, yet disruption prone. *Oral presentation.*
- 2022.** *Entomology 2022 Joint Annual Meeting.* Vancouver, Canada. Neotropical bumblebee gut microbiota are host conserved, yet disruption prone. *Oral presentation.*
- 2022.** *Sixth Congress of Computational Biology and Bioinformatics.* Cartagena, Colombia. La microbiota intestinal de abejorros del neotrópico es conservada, pero propensa a disrupción. *Oral presentation — Best Talk Award*
- 2021.** *ISME Latin American Congress.* Bogotá, Colombia. First exhaustive gut microbiota characterization of neotropical bees. *Online oral presentation.*
- 2020.** *Biological Sciences Conference, Universidad de los Andes.* Bogotá, Colombia. First exhaustive gut microbiota characterization of neotropical bees. *Online oral presentation.*
- 2019.** *Fourth International Forum on Computational Biology.* Bogotá, Colombia. ¿Cómo varía la microbiota intestinal en Abejorros del Neotrópico (*Bombus* sp.) en distintas altitudes?. *Poster presentation.*
- 2019.** *Fifth Conference on Computational Biology and Bioinformatics in Colombia.* Ibagué, Colombia. ¿Varía la microbiota intestinal en Abejorros del Neotrópico (*Bombus* sp.) en distintas altitudes?. *Oral presentation.*
- 2019.** *Research Symposium — UT SUPER Program.* Austin, Texas, USA. Does neotropical bumblebee (*Bombus*) gut microbiota vary between different altitudes?. *Oral and poster presentation.*

## GRANTS AND FELLOWSHIPS

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| <b>Miguel Velez Scholarship — University of California, Irvine</b> | 2024 |
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## WORKSHOPS & COURSES

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| <b>Workshop on Molecular Evolution — MBL, University of Chicago</b> | 2025 |
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**SKILLS**

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**Molecular Biology**

DNA extraction (tissue, blood, fecal, hair) · PCR, qPCR, gel electrophoresis · Amplicon and genomic library preparation (Ion Torrent, Illumina) · DNA quantification (Qubit, Bioanalyzer)

**Classical Microbiology & Lab Training**

BSL-2 laboratory practices · Bacterial isolation and culturing · Aseptic technique · Solid and liquid media preparation · Biochemical assays for bacterial identification · Antibiotic susceptibility testing · Phage inoculation assays

**Computational & Bioinformatics**

Genomic and metagenomic data analysis · Phylogenetic and evolutionary analysis · Statistical analysis and visualization in R · Bash scripting and Unix/Linux command-line workflows

**Scientific Communication & Visualization**

Scientific figure and presentation design · Adobe Illustrator, BioRender, Photoshop, Premiere, Reaper

**Field Biology**

Insect sampling (active and passive methods) · Insect identification and dissection · Specimen preservation and pinning

**SCIENCE COMMUNICATION & OUTREACH**

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**Creator & instructor — *Basis in Bioinformatics and Microbial Ecology workshop for Undergraduates* — Irvine, CA, USA. Jan 2026 - Present**

**Instructor — Clubes de Ciencia Colombia, ScienceLab & Science Minister of Colombia — Tuchín, Córdoba, Colombia. Nov 2025**

**Writer, Editor, and Translator — eTutorial at CABANA, Bogotá Mar 2022**

**Organizing Committee Member — ISME Latin American Congress, Bogotá Jul 2021**

**Led BEEMAS group, a discussion group for graduate and undergraduate students — Universidad de Los Andes Jan 2020 – Dec 2020**

**Volunteer — Progresía Fenicia (academic and social support for children) Jan 2017 – Dec 2017**

**LANGUAGES**

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**Spanish** (Native)   **English** (B2 — TOEFL certified)